

Lecture 2: Monopoly

ECON 7510

Cornell University

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Today's lecture

1. Monopoly pricing
2. Multi-product monopoly
3. Durable goods
4. Product quality

Monopoly pricing

Quantity choice under perfect competition

$$\max_q pq - C(q)$$

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FOC:

$$p = C'(q)$$

Price equals marginal cost. Zero profit on the marginal unit.

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Marginal revenue equals marginal cost.

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Marginal revenue equals marginal cost.

$$\begin{aligned}\Rightarrow p(q^m) &= C'(q^m) - \underbrace{p'(q^m)q^m}_{<0} \\ &> C'(q^m)\end{aligned}$$

Positive profit on the marginal unit.

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Positive profit on the marginal unit. *How much profit?*

Monopoly pricing

Equivalently, price choice: (Notation: $D = p^{-1}$.)

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$$p^m - C'(D(p^m)) = -\frac{D(p^m)}{D'(p^m)}$$

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“Lerner Index”: $L = -\frac{1}{\epsilon}$

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Question: What does this imply about perfectly elastic demand? Unit elastic demand?

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$$p^m = \left(\frac{1 + \epsilon}{\epsilon} \right) C'$$
$$p^m > 0 \Leftrightarrow \frac{1 + \epsilon}{\epsilon} > 0$$
$$\Leftrightarrow \epsilon < -1$$

Elastic part of the demand curve.

→ As long as demand is inelastic, $\frac{\partial \pi}{\partial p} > 0$, so increase price (i.e., decrease quantity) until you get to an elastic part of the demand curve.

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- Inefficiency? Yes, any deviation from $p = C'$ means quantity is inefficient.
- p^m is weakly increasing in marginal cost.

Proof: p^m is weakly increasing in marginal cost.

- Suppose $C_2'(q) > C_1'(q)$ for all $q > 0$.
- Let (p_1, q_1) and (p_2, q_2) denote the corresponding monopoly prices and quantities.
 - **Key idea:** Both (p_1, q_1) and (p_2, q_2) are points on the demand curve, so both feasible for both monopolists.

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which implies

$$\int_{q_2}^{q_1} \underbrace{[C_2'(x) - C_1'(x)]}_{>0 \ \forall x} dx \geq 0$$

so $q_1 \geq q_2$, which means $p_1 \leq p_2$.

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4. Suppose you know that firm is a monopolist and you know $p, \epsilon(p), c'(D(p))$. However, you observe that $p \neq \left(\frac{1+\epsilon}{\epsilon}\right) c'(D(p))$. What could explain this?

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4. Suppose you know that firm is a monopolist and you know $p, \epsilon(p), c'(D(p))$. However, you observe that $p \neq \left(\frac{1+\epsilon}{\epsilon}\right) c'(D(p))$. What could explain this?
 - Dynamic considerations: Reputation, resale market for durable goods, learning by doing, etc.
 - Multiproduct firm.

Multi-product monopoly

Setup

- Cost function: $C(q_1, \dots, q_n)$
- Demand: $D_1(\mathbf{p}), \dots, D_n(\mathbf{p})$

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Question: In what ways might the choice of price for one product involve considerations about other products?

First order condition

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Questions:

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Questions:

1. What's new here relative to single-product monopoly pricing?
2. What's the interpretation of each term?

Case 1: Separable costs, inseparable demands

Suppose $C(q_1, \dots, q_n) = \sum_i C_i(q_i)$. Can we derive an expression for L_i for each i ?

A definition that will prove useful:

$$\epsilon_{ij} = \frac{\partial D_j}{\partial p_i} \times \frac{p_i}{D_j}$$

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Case 2: Separable demands, inseparable costs

Suppose $q_i = D_i(p_i)$ for each i .

Example: Learning by doing in a two-period game.

- At $t = 1$, cost is $C_1(q_1)$.
- At $t = 2$, cost is $C_2(q_2, q_1)$, where $\partial C_2 / \partial q_1 \overset{?}{\geq} 0$.
- **Question:** What do you predict happens here?

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- FOC wrt p_1 :

$$D_1 + p_1 D_1' = C_1' D_1' + \delta (\partial C_2 / \partial q_1) D_1'$$
$$L_1 = -\frac{1}{\epsilon_1} + \delta \frac{1}{p_1} \frac{\partial C_2}{\partial q_1} < -\frac{1}{\epsilon_1}$$

Durable goods monopolist

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 - **Question:** What is the demand function?

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- **Question:** What might consumers do?

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- If the firm disappeared after $t = 1$, then at $t = 1$, the firm charges $p^m = \frac{1}{2}$ and sells $q^m = \frac{1}{2}$.
- If the monopolist (is surprised to find that she) continues to exist at $t = 2$, faces *residual demand* $D(p) = \frac{1}{2} - p$ and charges $p^m = \frac{1}{4}$ and sells $q^m = \frac{1}{4}$.
- **Question:** What might consumers do? If consumers anticipate that $p_2 = \frac{1}{4}$, they may decide to wait. And if the monopolist were not myopic, she'd anticipate this waiting....

Durable goods with commitment

- Suppose the monopolist can commit to a path of future prices.
- Can't do better than $p_t = p^m$ for all t . [Bulow (1982)]

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- **Question:** How likely is it that the monopolist actually has commitment power?

Coase Conjecture

Consider an extreme commitment problem. Suppose the monopolist has no commitment power, can set prices at $t = 0, \Delta T, 2\Delta T, 3\Delta T, \dots$, and consumers are fully rational and get gross utility θe^{-rt} if they purchase at t .

Coase Conjecture: Under some conditions the monopolist's profits go to zero as $\Delta T \rightarrow 0$.

Intuition:

- Suppose that $\theta \sim U(0, 1)$ and $c \in [0, 1]$
- Suppose that in the limiting time path of prices is not an immediate drop to zero.
- Hence, prices must be dropping linearly in T at some point and quantities must also be proportional to ΔT .
- Suppose the monopolist jumps ahead in its price sequence and charges $p_{r+\Delta T}$ instead of p_r .
- The gain from having all sales occur ΔT earlier is first order in ΔT .
- The loss from earning less on the sales at time t is second order in ΔT the price difference and quantity on which you get the lower price are both of order ΔT . Hence, it is better to jump ahead and cut prices faster.

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- The required conditions are actually somewhat narrow: Whether the result is true depends on whether the lower-bound of the value distribution is above or below c .
- Durable goods producers don't seem to earn zero profit. **Why? What was Coase missing?**
 - Delays/costs of changing prices (unlikely to be explanation in most contexts)
 - Inflows of new high willingness-to-pay customers
 - Per-period fixed costs
 - Strategic actions:
 - » Rent rather than sell (but can run into moral hazard or antitrust problems) (e.g., software)
 - » Most favored customer contracts or money back guarantee
 - » Destroy/limit ability to produce (e.g., art)
 - » Convince consumers that marginal cost is higher than it really is

Product quality

Monopoly and Product Quality

- Suppose the monopolist also chooses the quality $s \in \mathbb{R}$ of its good.
- Suppose it has a constant marginal cost $c(s)$ with $c'(s) > 0$, i.e. $C(q, s) = qc(s)$.
- Suppose a unit mass of consumers with types $\theta \sim U(0, 1)$ have unit demands: Utility from one unit is $v(s; \theta) - p$ where $v_s(s; \theta) > 0$ and $v_\theta(s; \theta) > 0$.

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The marginal cost of increasing s equals the marginal benefit to the marginal consumer.

Product Quality: Monopoly versus First-Best

Monopolist FOC: (Chooses optimal quality for the *marginal* consumer)

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Question: Does the monopolist choose higher or lower s than the social planner?

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Takeaway: Quality will almost certainly be distorted, but we have no general prediction on the direction of the distortion. (One of the key ambiguous theoretical results I referenced in Lecture 1.)